PHARMACOVIGILANCE // DATA SCIENCE REPORT

Drug Adverse Event Analysis and Risk Profiling

Predicting the seriousness of FDA-reported adverse drug events

800	8	0.78	68%
Reports Analyzed	Drugs Studied	Model ROC-AUC	Serious Rate

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Data source: FDA openFDA Adverse Event Reporting System (FAERS)

Executive Summary

Regulatory agencies such as the FDA receive millions of adverse event reports each year, and manually triaging which ones are clinically serious is slow and costly. This report analyzes 800 real-world adverse event reports pulled from the FDA's openFDA Adverse Event Reporting System (FAERS) across eight commonly prescribed drugs, with the goal of identifying which drugs, patient demographics, and reaction types are the strongest indicators of a serious outcome, and testing whether seriousness can be predicted from these features using machine learning.

800	68%	0.784	91%
Adverse Event Reports	Classified as Serious	Model ROC-AUC	Amoxicillin Seriousness Rate

The analysis followed a complete data science pipeline: data ingestion from the openFDA API, cleaning and encoding, exploratory analysis, feature engineering, hyperparameter tuning, and training of a Gradient Boosting classifier — achieving an ROC-AUC of 0.784, well above the 0.500 random baseline.

Methodology

Data was queried live from the openFDA Drug Adverse Event API for eight widely used drugs — Aspirin, Ibuprofen, Metformin, Lisinopril, Atorvastatin, Amoxicillin, Omeprazole, and Metoprolol — with 100 reports retrieved per drug (800 total). For each report, the patient's age, sex, primary reaction, seriousness classification, and reaction outcome were extracted.

The raw data was cleaned by converting FDA numeric codes into readable labels (e.g. sex, seriousness, outcome), validating patient ages to a plausible 0–120 range, and imputing the handful of missing ages with the per-drug median. No records were dropped during cleaning and all 800 reports were carried through to the final dataset.

For modeling, the cleaned data was transformed into a 31-feature matrix via one-hot encoding of drug, sex, age group (0–17, 18–39, 40–59, 60–74, 75+), the ten most frequent reaction types, and reaction outcome. A Gradient Boosting classifier was tuned with **GridSearchCV** across 24 hyperparameter combinations using stratified 5-fold cross-validation, optimizing for ROC-AUC.

Exploratory Data Analysis

Seriousness Rate by Drug

Seriousness rates varied sharply across drugs, ranging from 46% for Metoprolol to 91% for Amoxicillin. Amoxicillin, Atorvastatin, and Ibuprofen stood out as the highest-risk drugs in this sample.

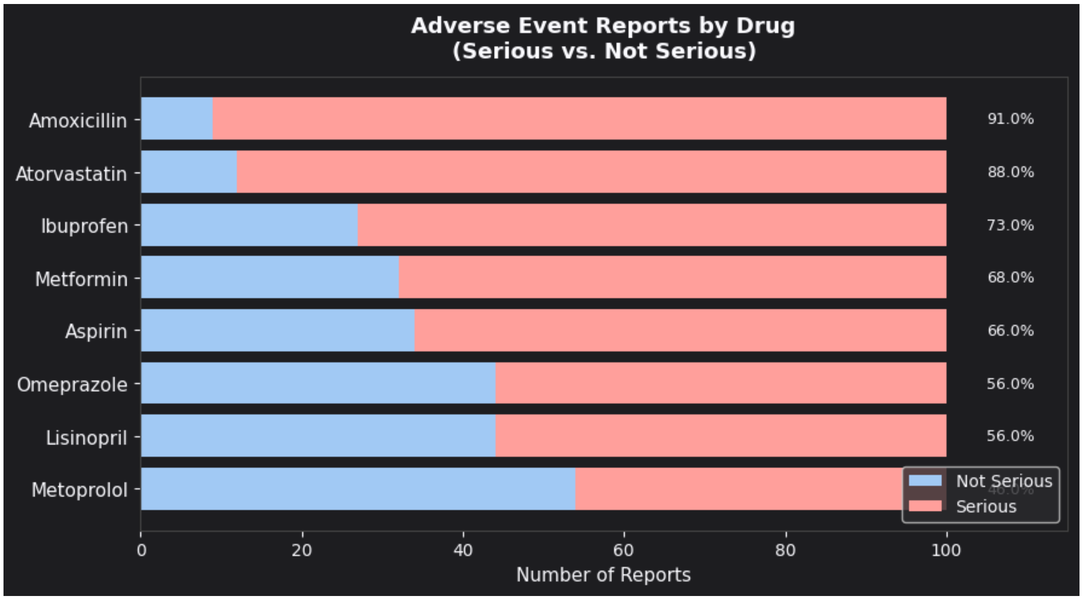

Figure 1. Adverse event reports by drug, split by seriousness, with the seriousness rate labeled.

Drug	Total Reports	Serious	Seriousness Rate
Amoxicillin	100	91	91.0%
Atorvastatin	100	88	88.0%
Ibuprofen	100	73	73.0%
Metformin	100	68	68.0%
Aspirin	100	66	66.0%
Lisinopril	100	56	56.0%
Omeprazole	100	56	56.0%
Metoprolol	100	46	46.0%

Most Frequently Reported Reactions

Type 2 Diabetes Mellitus was the single most common reaction reported (39 cases), followed by Gastrointestinal Haemorrhage (19) and Diarrhoea (18). Several of the top reactions, including Haemorrhage, Cerebrovascular Accident, and Death, are inherently severe events.

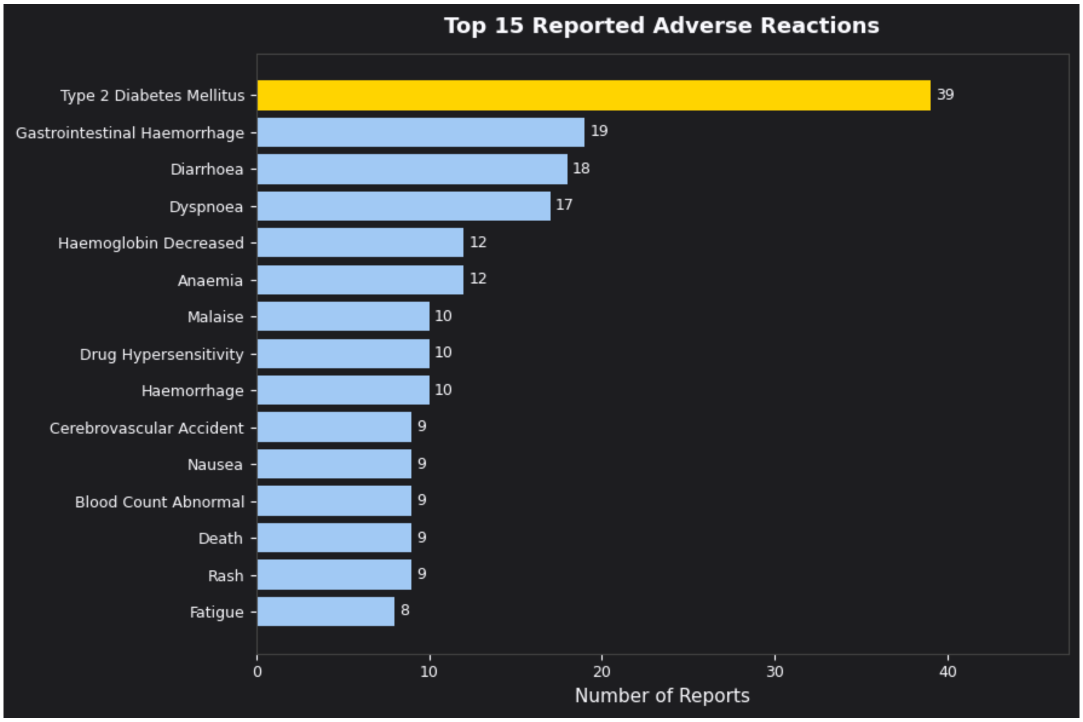

Figure 2. Top 15 reported adverse reactions by report count.

Patient Age and Seriousness

The average patient age was 61.7 years. Somewhat counter-intuitively, patients with serious events skewed slightly younger on average (mean 60.1) than those with non-serious events (mean 65.0), though both groups were concentrated in the 55–71 age range.

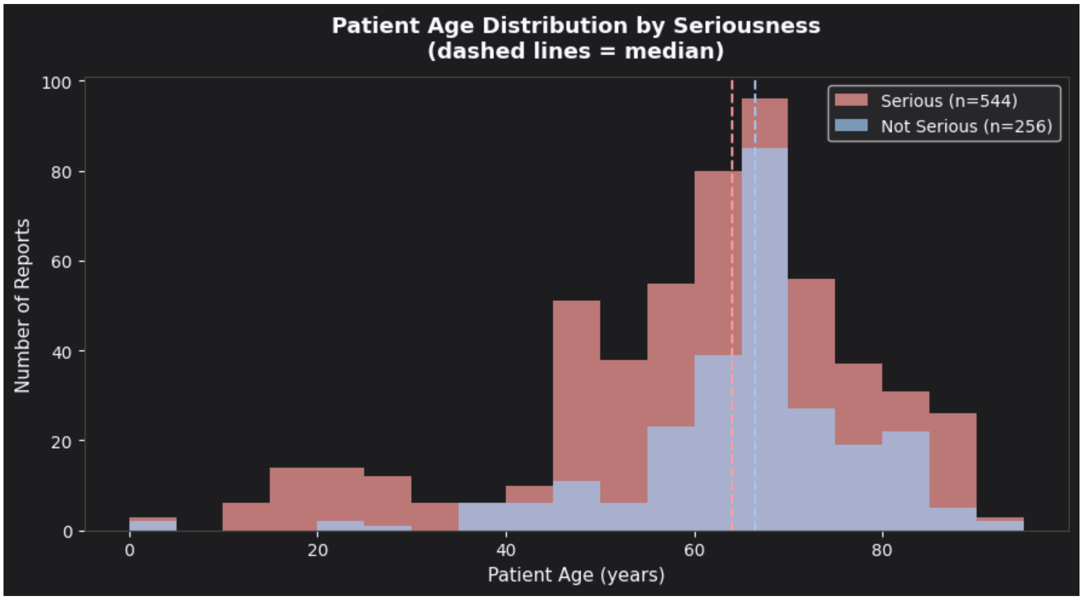

Figure 3. Patient age distribution by seriousness, with medians marked.

Reaction Outcomes

Outcome was unknown/unreported for roughly half of all cases (399 of 800). Among reports with a known outcome, 42 resulted in death and 13 in recovery with lasting sequelae — both outcome categories were classified as serious 100% of the time, as expected.

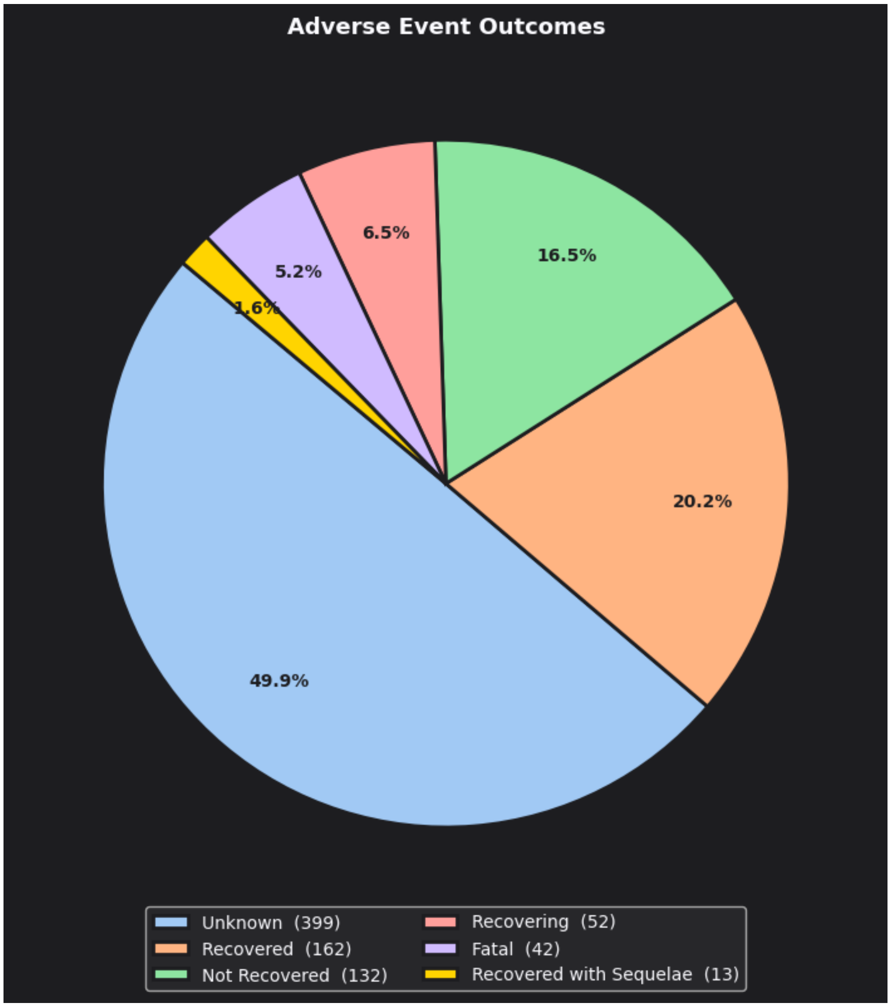

Figure 4. Distribution of reaction outcomes across all reports.

Demographic Breakdown

Female patients accounted for more reports than male patients (441 vs. 332, with 27 unrecorded), but seriousness rates were nearly identical between sexes — 68.7% for female patients versus 67.5% for male patients — indicating that sex alone is not a strong differentiator of risk in this dataset.

Predictive Modeling

A Gradient Boosting classifier was trained to predict whether a report would be serious, using the 31-feature encoded dataset (640 training / 160 test reports, stratified 80/20 split). Hyperparameters were selected via grid search over 24 combinations of tree depth, learning rate, number of estimators, and minimum samples per leaf.

Hyperparameter Tuning

Model	Train AUC	Test AUC	Accuracy	Overfit Gap
Baseline GBM (untuned)	0.880	0.803	73.8%	0.077
Tuned GBM (GridSearchCV)	0.862	0.784	71.9%	0.078

Best parameters: learning_rate = 0.05, max_depth = 3, min_samples_leaf = 10, n_estimators = 200 (best inner cross-validated ROC-AUC: 0.772). The tuned model's held-out performance was close to the untuned baseline, suggesting the baseline configuration was already reasonably well suited to this dataset; the final reported model uses the grid-search-selected parameters for consistency and robustness.

Final Model Performance

71.9%	0.784	85%	43%
Accuracy	ROC-AUC	Serious Recall	Not-Serious Recall

The model correctly identifies 85% of truly serious events (recall), which is valuable in a triage setting where missing a serious case is costlier than a false alarm. Precision and recall are lower for the minority "Not Serious" class (43% recall), a common effect of class imbalance — 68% of the dataset is labeled serious.

	Precision	Recall	F1-Score	Support
Not Serious	0.58	0.43	0.49	51
Serious	0.76	0.85	0.81	109
Accuracy			0.72	160

Feature Importance

The strongest predictors of a serious outcome were drug identity (Amoxicillin, Atorvastatin), a fatal outcome flag, and a Type 2 Diabetes Mellitus reaction — consistent with the patterns seen in the exploratory analysis.

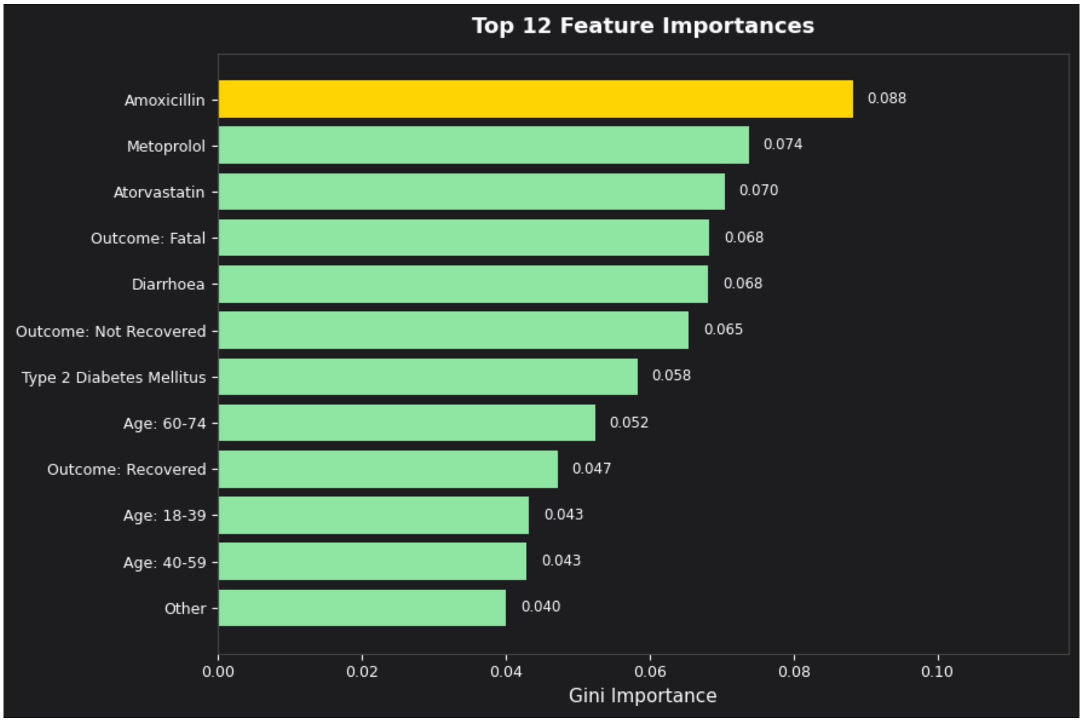

Figure 5. Top 12 features by Gini importance in the tuned Gradient Boosting model.

ROC Curve

The ROC curve confirms the model's discriminative power sits well above a random classifier, with an AUC of 0.784 on the held-out test set.

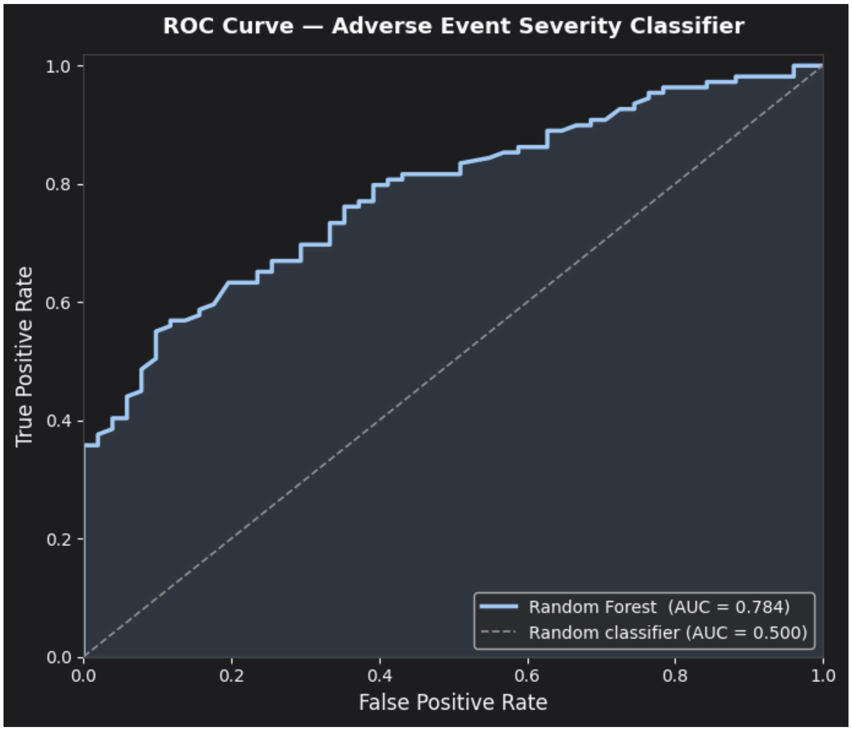

Figure 6. ROC curve for the adverse event severity classifier.

Key Findings

Adverse Event Landscape

- Amoxicillin had the highest seriousness rate at 91%, followed by Atorvastatin (88%) and Ibuprofen (73%).
- The most commonly reported reactions were Type 2 Diabetes Mellitus, Gastrointestinal Haemorrhage, and Diarrhoea.
- 68% of all 800 events were classified as serious by reporters.
- 42 reports resulted in death — all classified as serious (100%).

Patient Demographics

- Mean patient age was 61.7 years (serious: 60.1 | not serious: 65.0).
- Female patients outnumbered male patients in the sample (441 vs. 332), with comparable seriousness rates (~68% for both).
- Most reports came from patients aged 52–71 (interquartile range).

Machine Learning Model

- A tuned Gradient Boosting Classifier reached an ROC-AUC of 0.784, indicating meaningful predictive power well above chance (0.500).
- Top predictors were drug type (Amoxicillin, Atorvastatin), a fatal reaction outcome, and a Type 2 Diabetes Mellitus reaction.
- Moderate accuracy (71.9%) reflects the real-world complexity and noise typical of adverse event reporting; a larger dataset would likely improve results further.

Conclusion

This analysis demonstrates that a meaningful signal for adverse event seriousness can be recovered from readily available FAERS fields — drug identity, patient demographics, reaction type, and outcome — using a standard gradient boosting approach. While the current model is not accurate enough to replace expert clinical triage, an ROC-AUC of 0.78 suggests this type of model could reasonably assist in prioritizing which reports warrant closer human review first, particularly given its strong recall (85%) on serious cases. Expanding the drug list, increasing the sample size per drug, and incorporating richer clinical features (e.g. concomitant medications, dosage, comorbidities) are natural next steps to improve performance further.

Data source: FDA openFDA Drug Adverse Event API (api.fda.gov). This report is for portfolio and educational purposes and does not constitute medical or regulatory guidance.